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第二章 函数的导数(Derivative)

2.1 导数的定义(The Definition of Derivative)

Multiple Choice Questions

$\frac{1}{3}$ 1, 2

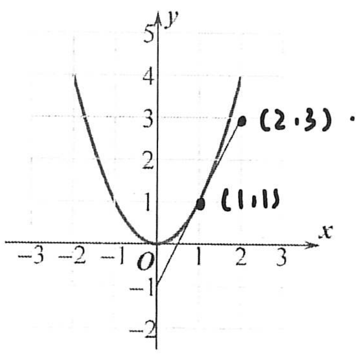
C 1. Which of the following values is the average rate of change of $f(x)=\sqrt{x+2}$ over the interval $[-1,2]$?

(A) -4 (B) $-\frac{1}{3}$ (C) $\frac{1}{3}$ (D) 3

C 2. Let $f(x)=4-3x$. Which of the following is equal to $f'(0)$?

(A) 4 (B) -4 (C) -3 (D) 4

D 3. The graph of f and its tangent line at $x=1$ is shown below, find $f'(1)=($ $)$.



(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2

$\frac{k(-2)^3+4-(-k+2)}{1}=3$

C 4. Let $f(x)=kx^3-2x$, $(k\neq 0)$. The average rate of change of $f(x)$ over the interval $[-2,-1]$ is 3, find k.

(A) $\frac{5}{7}$ (B) $\frac{1}{7}$ (C) $-\frac{1}{7}$ (D) $-\frac{5}{7}$

$-8k+4+k-2=3$

$-7k+2=3$

$-7k=1$

$k=-\frac{1}{7}$

C 5. **Caculator** Let $y=\tan e^x-x^2+1$. Find $\frac{dy}{dx}\Big|_{x=0.5}=$

(A) -12.056 (B) 0 (C) 271.066 (D) 273.066

C 6. **Caculator** The graph of $y=\sqrt{x^2+1}-3x$ crosses the x-axis at one point in the interval $[0,1]$. What is the slope of the graph at this point?

(A) 0.354 (B) -0.001 (C) -2.666 (D) 0

$\sqrt{x^2+1}=3x$

$x^2+1=6x^2$

$5x^2=1$

$x^2=\frac{1}{5}$

$x=\pm\frac{1}{\sqrt{5}}=\pm\frac{\sqrt{5}}{5}$

$\because 0<x<1$

$\therefore x=\frac{\sqrt{5}}{5}$

A 7. The population of insects at time t days is modeled by the function P with first derivative $P'(t)=0.1t^2+10t+120$. Which of the following statements is true?

(A) At time $t=10$, the population of insects is 230.

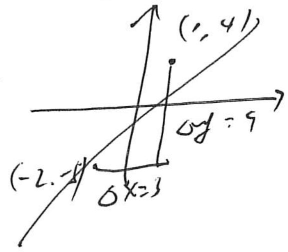
(B) At time $t=10$, the population of insects is increasing at a rate of 230 insects per day.

(C) At time $t=10$, the population of insects is decreasing at a rate of 230 insects per day.

(D) At time $t=10$, the rate of change of the number of insects is increasing at a rate of 230 insects per day.

A 8. If the line tangent to the graph of a function g at the point $(1,4)$ passes through the point$(-2,-5)$, what is the slope of this line?

(A) 3 (B) -3 (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$



Free Response Questions

9. Let $f(x)=x^2-2x$

(a) Find the average rate of change of f over $[1,3]$.

(b) Find $\frac{f(2+h)-f(2)}{h}$.

(c) Find the instantaneous rate of change of f at $x=2$.

a: $\frac{f(1)-f(3)}{1-3} = \frac{-1-3}{-2} = 2$

b: $\lim_{h\to 0}\frac{(2+h)^2-2(2+h)-4+4}{h} = \lim_{h\to 0}\frac{4+h^2+4h-4-2h}{h} = \lim_{h\to 0} h+4-2 = 2$

10. Use the definition of derivative to find the derivative of $f(x)=x^3$, then find $f'(2)$.

$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{(x+h)^3-x^3}{h} = \lim_{h\to 0}\frac{x^3+3xh^2+3hx^2+h^3-x^3}{h}$

$=\lim_{h\to 0} 3xh+3x^2+h^3$

$=3x^2$